Bioelectronics

Timo Streule, 13.07.2019
tstreule@student.ethz.ch

General

Formeln

Circle: $U_{\text{circle}} = 2\pi r$, $A_{\text{circle}} = \pi r^2$

Constants:

$h = 6.626\,\text{E}{-34}$ Js $= 4.135\,\text{E}{-15}$ eV s, $\hbar = \frac{h}{2\pi}$, $hc = 1.986\,\text{E}{-25}$ Jm

$\epsilon_0 = 8.85\,\text{E}{-5}$ $^{\text{As}}/_{\text{Vm}}$

$\mu_0 = 4\pi\,\text{E}{-7}$ $^{\text{N}}/_{\text{A}^2}$

$k_B = 1.38\,\text{E}{-23}$ $^{\text{J}}/_{\text{K}} = 8.617\,\text{E}{-5}$ $^{\text{eV}}/_{\text{K}}$

$q = 1.602\,\text{E}{-19}$ C, $m_e = 9.109\,\text{E}{-31}$ kg, $m_p = 1.672\,\text{E}{-27}$ kg

$F = 96485\,^{\text{C}}/_{\text{mol}}$ (Faraday), $R = N_A k_B = 8.314\,^{\text{J}}/_{\text{mol K}}$ (Ideal gas constant)

$N_A = 6.022\,\text{E}23$ $^{\text{particles}}/_{\text{mol}}$ $0\,°\text{C} = 273.15\,\text{K}$

Einheiten

Druck	$1\,\text{Pa} = 1\,^{\text{N}}/_{\text{m}^2} = 1\,^{\text{J}}/_{\text{m}^3} = 10\,^{\text{g}}/_{\text{cm·s}^2}$
Induktivität	$1\,\text{H} = 1\,^{\text{Vs}}/_{\text{A}} = 1\,\Omega\,\text{s}$
Power	$1\,\text{W} = 1\,^{\text{J}}/_{\text{s}} = 1\,\text{VA}$
electron volt	$1\,\text{eV} = 1.602\,\text{E}{-19}$ J $= 23.06\,^{\text{kcal}}/_{\text{mol}}$
Charge	$1\,\text{C} = 1\,\text{A s}$
Energy	$1\,\text{J} = 1\,^{\text{kg m}^2}/_{\text{s}^2} = 1\,\text{Nm} = 1\,\text{VAs} = 1\,\text{CV} = 1\,\text{Ws}$

Good to know

Power in dB	$10\log_{10}\frac{I}{I_0}$
Intensity	$I = \frac{\text{avg. Power } (P)}{\text{area } (A)}\left[\frac{\text{W}}{\text{m}^2}\right]$
avg. Power	$P = \frac{\text{avg. Work in cycle } (\overline{W})}{\text{cycle } (T)}\left[\text{W} = \frac{\text{kg m}^2}{\text{s}^3}\right]$
(avg.) Work	$\overline{W} = \int_{\text{cycle}} \frac{1}{\text{cycle}(T)} \cdot \vec{F} \cdot \vec{x}\,\text{d}t\,[\text{J} = \text{Ws}]$

mass m vibrates with an amplitude a along x-axis:
$\vec{x}(t) = a\sin(\omega t)\cdot\vec{e}_x$, $\omega = 2\pi f = 2\pi/T$, $\vec{F}(t) = m\,\ddot{\vec{x}}(t)$

NuS

Induktivität	$u(t) = L\frac{\text{d}i_L}{\text{d}t} \circ\!\!-\!\!\bullet\ U(s) = sL\,I_L(s)$
Konduktivität	$i(t) = C\frac{\text{d}u_C}{\text{d}t} \circ\!\!-\!\!\bullet\ U(s) = \frac{1}{sC}U_C(s)$
Transformator	$u_1 = L_1\frac{\text{d}i_1}{\text{d}t} - M\frac{\text{d}i_2}{\text{d}t}$, M: mutual inductance
LCR-Schwingkreis	$\omega_0 = 2\pi f_0 = 1/\sqrt{LC}$

Laplace

$\lambda\,u(t) + \mu\,v(t)$	$\lambda\,U(s) + \mu\,V(s)$
$u(at)$, $a>0$	$\frac{1}{a}\,U(\frac{s}{a})$
$u(t-t_0)$	$\text{e}^{-st_0}\,U(s)$
$\text{e}^{-at}\,u(t)$	$U(s+a)$
$(-t)^n u(t)$	$U^{(n)}(s)$
$u^{(n)}(t)$	$s^n U(s) - \ldots - u^{(n-1)}(0)$

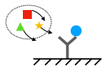
t^n	$\frac{n!}{s^{n+1}}$	$t^n\text{e}^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
e^{-at}	$\frac{1}{s+a}$	$\sin(at)$	$\frac{a}{s^2+a^2}$
$t\text{e}^{-at}$	$\frac{1}{(s+a)^2}$	$\cos(at)$	$\frac{s}{s^2+a^2}$

1 Biosensors

1.1 Label assay vs. label-free

Label assay (sandwich): 

Label-free assay (OWLS): 

non-specific binding: 

other interactions that could happen, except the angle binding.

1.2 Rules of Chemistry and Physics

Bracket notation indicates concentration $\text{A} + \text{B} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \text{AR}$

equilibrium constant $K = \frac{k_{-1}}{k_1} = \frac{[A][R]}{[AR]}$

affinity constant $K_a = 1/K$

cont. flowing cell $[AR] \sim q * \text{signal}$, $q = const$

$[R] \sim R_0 =$ initial receptor density

1.3 Sensitivity and Specificity

Sensitivity: true positive rate (% of correctly identified +)

Specificity: true negative rate (% of correctly identified −)

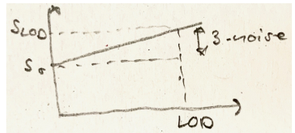
Compensate Sensitivity through LOD.

1.3.1 Limit of Detection (LOD) (Sensitivity)

Limitation by non-specific binding (NSB) → noise at zero analyte

LOD $LOD = 3\cdot\text{noise}/\frac{\text{d}S}{\text{d}[A]}$

$S_{LOD} = S_0 + 3\cdot\text{noise}$



LOD for intensity/signal $LOD_{NSB} = \langle I_{NSB}\rangle + 3\,\sigma(I_{NSB})$

Lowest detectable intensity $\frac{\langle I_{POI}\rangle}{LOD_{NSB}} = 1$ (POI: proteins of interest)

Detectable #proteins $N\# = \frac{\Gamma}{m_{\text{protein}}}$ $[\Gamma] = {}^{\text{pg}}/_{\text{mm}^2}$ detection limit

2 Optical Microscopy

2.1 Reflection and Refraction

Law of Reflection $\theta_{\text{inc}} = \theta_2$

Law of Refraction $\frac{n_{\text{inc}}}{n_2} = \frac{\lambda_0/\lambda_{\text{inc}}}{\lambda_0/\lambda_2} = \frac{\sin\theta_2}{\sin\theta_{\text{inc}}}$ for $\theta_{\text{inc}} < \theta_c$

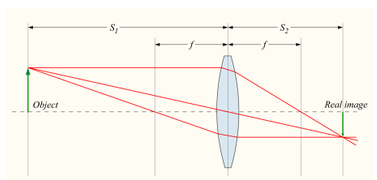
Total reflection $\sin\theta_c = n_2/n_{\text{inc}}$ **always total** if $n_{\text{inc}} < n_2$

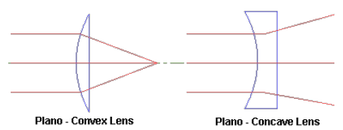
Paraxial approx. $\theta \simeq \sin\theta \simeq \tan\theta$, $\theta \ll 1$

Thin lens approx. $R \ll S_o, S_i$

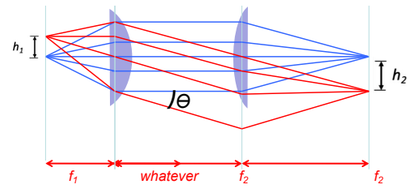
Lens makers formula $\frac{1}{S_o} + \frac{1}{S_i} = \frac{1}{f} = \frac{n_{\text{lens}} - n}{n}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

2.2 Ray Tracing





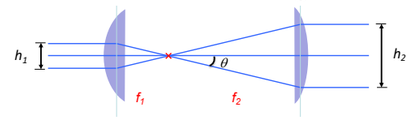
2.2.1 Simple Microscope



Magnification $M = h_2/h_1 \overset{*}{=} f_2/f_1$ * since $h_1 = f_1\sin\theta$

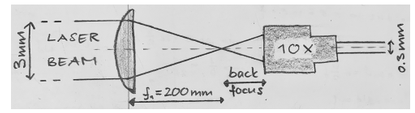
2.2.2 Beam Expander

for light source

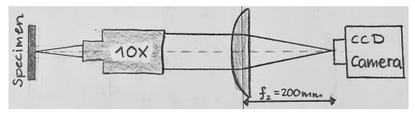


2.2.3 Simple "modern" Microscope

(uniform!) Illumination: Camera: $(M = f_{tube}/f_{obj})$



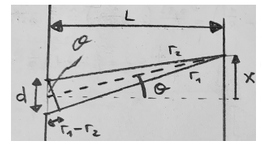
(Objective lens)



(Tube lens)

2.3 Fundamental Limits of Lenses & Resulution

Due to a **finite aperture** will points presented as the PSF (point-spread).



Rayleigh criterion First zero occurs at: $\theta \overset{(\theta\ll1)}{\simeq} \sin\theta \simeq 1.22\frac{\lambda_0}{d}$

d: radius of the aperture

- $\sin\theta \simeq \tan\theta = x/L$
- $r_1 - r_2 = d\cdot x/L$

maximum of the one is on the minimum of the other curve (FWHM)

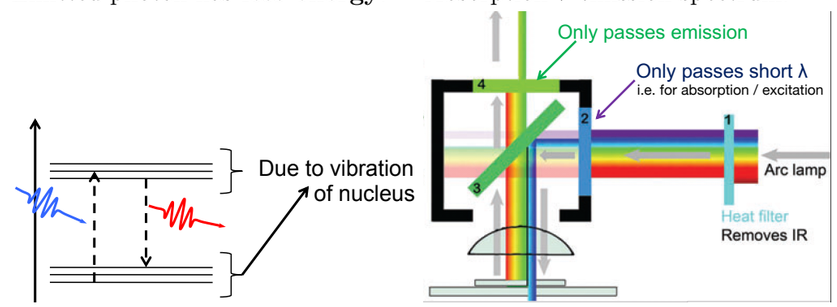
Resolution $\Delta r \simeq f\sin\theta \times M \simeq 1.22\frac{\lambda_0 f}{D} \times M = 0.61\frac{\lambda_0 n}{\text{NA}} \times M$

Numerical Aperture $\text{NA} = n\sin\theta \simeq n\,\frac{D}{2f}$ n: refractive index,

D: aperture size of lens, f: focal length of lens

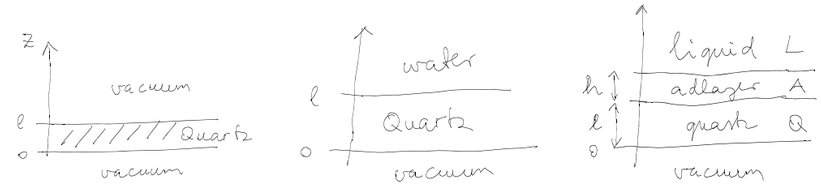
2.4 Fluorescence Microscopy

Emitted photon has **less energy**. → Absorption ≠ emission spectrum.



3 Mechanical Sensors

3.1 Viscoelastic media



Remark: Calculate *always* at first in $^{\text{rad}}/_{\text{s}} \xrightarrow{\times 1/2\pi}$ Hz.

3.1.1 Crystal with thickness l

Resonance frequency $\omega_n = n \cdot \omega_0 = n \cdot \frac{\pi}{l}\sqrt{\frac{\mu_Q}{\rho_Q}}$ $[\mathrm{rad/s}]$

char. wavelength $\lambda = 2l/n$

3.1.2 Elastic plate in water

Resonance frequency $\omega_n = n\omega_0 + \Delta\omega_n$

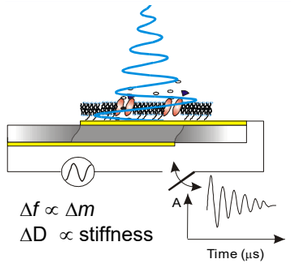
frequency shift $\Delta\omega_n = -\sqrt{n}\sqrt{\frac{\rho_L \eta_L \omega_0}{2}}\frac{1}{l\,\rho_Q}$, η_L: viscosity

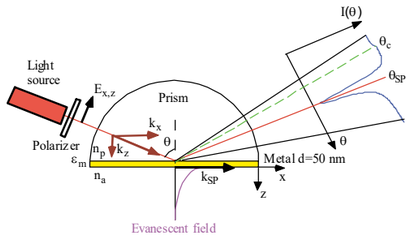
3.1.3 Elastic plate in water with adlayer

In fact is the *resonance frequency* dependent on the adlayer height.
But if it is "enough" small, the Sauerbrey approximation is sufficient.

Sauerbrey eq. $\Delta f_n = 2\pi\omega_n = -n\frac{\Delta m}{C}$ $\quad C = \frac{\sqrt{\rho_Q \mu_Q}}{2f_0^2} = 17.7\,\mathrm{ng/cm^2Hz}$

3.2 QCM-D vs. SPR techniques





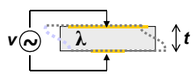
$\Delta m = -\frac{C}{n}\Delta f_n$ $\qquad$ $\Delta m = d\,\frac{n_{\mathrm{protein}} - n_{\mathrm{buffer}}}{\mathrm{d}n/\mathrm{d}c}$

$\mathrm{LOD_{QCM}} = C \cdot \delta f \simeq 1\,\mathrm{ng/cm^2}$ $\qquad$ $\mathrm{LOD_{SPR}} = C \cdot \delta\theta \simeq 0.1\,\mathrm{ng/cm^2}$

3.3 QCM: Quartz Crystal Microbalance

Resonance condition $f = \frac{nv}{\lambda} = \frac{nv}{2t}$

Dissipation $D = \frac{1}{\pi f \tau}$, τ: time until $\frac{U_{\max}}{\mathrm{e}}$



Modeling of the QCM-D response in air / aqueous solution:



η: viscosity ($= \frac{G''}{\omega}$), μ: elasticity ($= G'$),
ρ: density, d: thickness

3.4 Strain Gauge

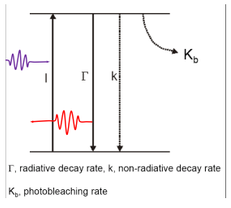
Resistive strain g. $\frac{\Delta R}{R} = k\,\frac{\Delta l}{l} = k\,\epsilon$, ϵ: Strain, k: Gauge factor

Capacitive strain g. $C = \epsilon_0\epsilon_r\,\frac{A}{d}$, (displacement: $d \to d + \Delta d$)

4 Fluorescent Probes

4.1 Fluorescence Statistics

Photobleaching: A fluorescent molecule can emit a limited #photons by excitation before it irreversibly converts to a non-fluorescent molecule.



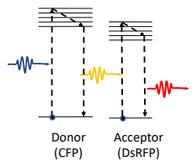
Quant. Yield $Q = \frac{\Gamma}{\Gamma + k + K_b} \sim 0 - 98\%$ (Efficiency)

Fluorescence **Lifetime** $\tau = \frac{1}{\Gamma + k + K_b} \sim 1\,\mathrm{ns}$

Fluorescence emission is a statistical process that is characterized by exponential decays.

Molecules in *excited* state $\frac{\mathrm{d}N_e}{\mathrm{d}t} = -\left(\frac{1}{\tau_1} + \ldots + \frac{1}{\tau_n}\right)N_e$

4.2 FRET – Fluorescence Resonance Energy Transfer



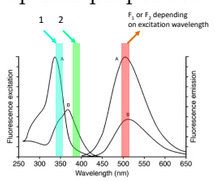
Seeing a different $\lambda_{\mathrm{emission}}$ tells us that two molecules are close.

Efficiency $E = \frac{R_0^2}{R_0^2 + r^6} = 1 - \frac{\tau_{\mathrm{DA}}}{\tau_{\mathrm{D}}} = 1 - \frac{I_{\mathrm{DA}}}{I_{\mathrm{D}}}$ where

$\tau_{\mathrm{D(A)}}$, $I_{\mathrm{D(A)}}$ Lifetime/Intensity of donor emission in the absence/presence of acceptor.

4.3 Calcium Imaging → too slow for t dependency measurements

Calcium ion cannot be visualized/tagged directly: → Design molecules with optical properties that change upon calcium binding.



Single & dual wavelength measurements:

Concentration $[\mathrm{Ca^{2+}}]_i = K_{\mathrm{d,eff}}\frac{R - R_{\min}}{R_{\max} - R}$ where

for single $R \equiv F$ and $K_{\mathrm{d,eff}} = \frac{[\mathrm{Ca^{2+}}]_i \times (F_{\max} - F)}{F - F_{\min}}$

for dual. $R = F_1/F_2$ (F: fluorescence)

The binding of $\mathrm{Ca^{2+}}$ leads to...

- change in fluorescence **intensity** but not wavelength change
- a **shift** in excitation (and/or emission) peaks ("dual wavelength")
- changes in fluorescent resonance energy transfer (**FRET**) & life time

4.3.1 Delivery of Calcium Indicators

Loading cells Once the molecule got cleaved (spalten), it cannot go out and gets fluorescent.

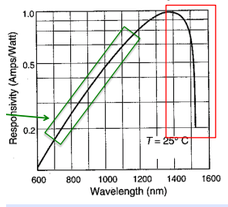
Introduce **fluoresc. proteins** or (natural) **FRET proteins** Proteins changes emission rate when $\mathrm{Ca^{2+}}$ binds. The relative change ($\Delta F/F$) in the fluorescence emission of EGFP can then be directly measured.

Note: The dye responds *fast*, but it is *slow* to recover. Calcium dissociation rate (i.e. recovery) depends on the dye's affinity for Calcium.

5 Detection and Noise

5.1 Signal

For statistically independent, "uncorrelated", events we use



Poisson $P_{\overline{n}}(n) = \frac{\overline{n}^n}{n!}\mathrm{e}^{-\overline{n}}$ $\overline{n}$: mean, $\sigma_n = \sqrt{\overline{n}}$: var

Signal current $\langle I_{\mathrm{sig}} \rangle = \eta q \overline{n}/\Delta t$

Responsivity $\langle I_{\mathrm{sig}} \rangle / P = \eta q \lambda/(hc)$

Cutoff λ $\lambda(\mu\mathrm{m}) = 1.24/E(\mathrm{eV})$

5.1.1 Fundamental Noise Sources

Noise

shot $N_{\mathrm{s}} = 2R\eta q B \langle I_{\mathrm{sig}} \rangle = \langle I^2_{\mathrm{shot}} \rangle R$, $2B \simeq 1/\Delta t$

dark $N_{\mathrm{d}} = 2R\eta q B \langle I_{\mathrm{dark}} \rangle$

thermal $N_{\mathrm{J}} = \langle I^2_{\mathrm{johnson}} \rangle R = \sqrt{\frac{k_{\mathrm{B}} T B}{R}}^2 R$

readout $N_{\mathrm{r}} = N_{\mathrm{J}} + N_{\mathrm{amplifier}}$

$N_{\mathrm{tot}} = N_{\mathrm{s}} + N_{\mathrm{d}} + N_{\mathrm{r}}$ $\quad \sigma^2_{\mathrm{tot}} = \sum \sigma_i^2$

SNR $\mathrm{SNR_{curr}} = \frac{\eta \cdot N_\gamma}{\sqrt{\eta N_\gamma + N_d + N_r}}$ $\quad \eta = \mathrm{QE} \cdot absorb$

for large numbers $\mathrm{SNR_{shot}} = \frac{\overline{n}}{\sqrt{\overline{n}}} = \sqrt{\overline{n}} \mathrel{\widehat{=}} \eta N_\gamma$

5.2 (Optical) Detectors

Photoel. effect $E_{\mathrm{ph}} = hf = hc/\lambda = \phi + E_{\mathrm{kin}}$

incident $E_{\mathrm{ph}} = E_{\mathrm{binding}} + E_{\mathrm{kin}}$ of ejected el.

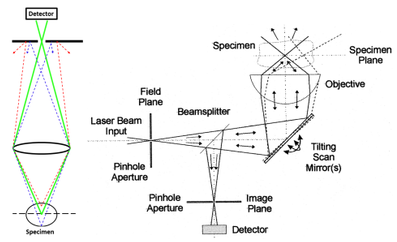
Photo mult. (PMT) $I = \alpha\,(S \cdot E_{\mathrm{ph}}\,n/\Delta t)$, S: sensitivity, $N_{\mathrm{d}} \uparrow$

Choose **binning mode** (2x2, 4x4, ...) such that the pixel size of the camera fits the best the maximum pixel size.

max. pixel size $\max = 0.3\frac{\lambda}{\mathrm{NA}} \times M$

5.3 Imaging Deep Tissues

Beer-Lambert's law $I = I_0\,\mathrm{e}^{-\mu x}$

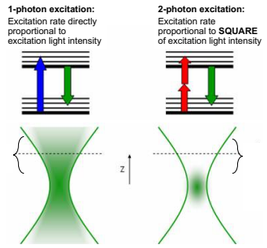


1st Pinhole by illumination:
→ focus the illum. to a small spot
2nd Pinhole by detector:
→ reject out of focus light
→ collect light to PMT

Signal and Noise (**Pinhole size**):
- more out of focus light → blurry
- less in-focus light → SNR↓

Choose **pinhole size** $\sim\Delta r = 0.61\frac{\lambda}{\mathrm{NA}} \times M$

5.3.1 2ph-Excitation



\+ wavelength↑ → less absorbing/scattering
\+ photons need to coincide in space and time to excite
\+ No pinhole by detector for 2ph: Collect all the light
– low probability of occurrence i.e. inefficient
– ultra-short pulse light source needed

6 Optical Biosensors

6.1 EM

Relations:

$c_0 = \frac{1}{\sqrt{\epsilon_0\mu_0}}$ $\quad k_0 = \frac{2\pi}{\lambda_0} = \frac{\omega}{c_0} = \omega\sqrt{\epsilon_0\mu_0}$ $\quad k = \frac{2\pi n}{\lambda}$ $\quad n = \frac{c_0}{c} = \sqrt{\epsilon_r\mu_r} \simeq \sqrt{\epsilon_r}$

Dispersion relation $k^2 = \epsilon\omega^2 = \epsilon k_0^2 c_0^2 = k_0 \frac{\epsilon}{\epsilon_0} = k_0^2 \epsilon_r = (k_0 n)^2$

Plane waves $\vec{E}(\vec{r}, t) = \vec{E}_0\,\mathrm{e}^{i(\vec{k}\vec{r} - \omega t)}$

whereby $\partial_t \mathrel{\widehat{=}} -i\omega$, $\partial_x \mathrel{\widehat{=}} ik_x$, $\partial_z \mathrel{\widehat{=}} ik_z$, $\partial_y \mathrel{\widehat{=}} 0$ (infinite extent)

Depth of penetration $d_p = \frac{\lambda}{4\pi}\frac{1}{\sqrt{n_{\mathrm{inc}}^2 \sin^2\theta - n_2^2}}$ (about 500 nm)

6.2 Evanescent Field Techniques

Recap Always **total reflection** for $n_{\mathrm{inc}} > n_2$

Evanescence Negligible change of sensitivity compared to the size of the antibodies

6.2.1 SPR – Surface Plasmon Resonance

Plasmon quantum of an electron density wave in a metal

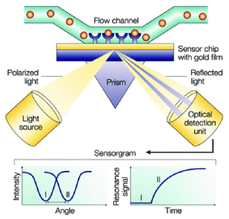
Plasmon Polariton mixture of photon (diel.) and el. dens. wave (met.)

SPP field components point in direction of propagation

Dispersion relation $k_{z,i}^2 = k_0^2\epsilon_i - \beta^2$ $\quad i = d, m$

Momentum of incoming wave $\beta := k_x = \underbrace{\frac{\omega}{c}}_{k_0}\underbrace{\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}}_{N}$

where $N = \sqrt{\frac{1}{n_c^2} + \frac{1}{\epsilon_m}}$ effective refr. index of SP

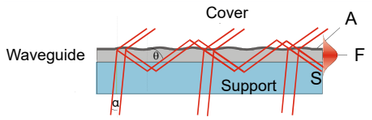


Launch a SPP: $0 \le k_{\parallel,\mathrm{inc}} \le k_0\, n_{\mathrm{inc}}$

must ensure $\beta \overset{!}{=} k_{\parallel,\mathrm{inc}} = k_0\, n_{\mathrm{inc}}\, \sin\theta$ $\quad \theta \in [0, \frac{\pi}{2}]$

$\Longrightarrow$ $n_{\mathrm{inc}} \ge n\sqrt{\frac{\epsilon_m}{\epsilon_m + n_c^2}}$ $\quad \frac{\Delta\theta}{\Delta N} \triangleq \frac{\partial\theta}{\partial N} = \frac{1}{n_{\mathrm{inc}}\cos\theta} \simeq \frac{1}{n_{\mathrm{inc}}}$

6.2.2 OWLS – Optical Waveguide Lightmode Spectroscopy



$N \coloneqq n_F \sin\theta = n_{\mathrm{air}} \sin\alpha + \frac{l\lambda}{\Lambda}$

Penetration depth $\sigma \sim \frac{\lambda}{2\pi}\frac{1}{\sqrt{N^2 - n_c^2}}$

Waves have to be in phase (constr. interference) → extremely sensitive

constr. interference $0 = 2\pi m \overset{!}{=} \phi_F + \phi_{FS} + \phi_{FAC}$ $\quad \phi$: phase shifts

Ansatz for *3 layer model*

$$\begin{cases} \text{Cover:} & C\,\mathrm{e}^{-|k_{z,C}|\,(z-d_F/2)} \\ \text{Waveguide:} & B\,\mathrm{e}^{ik_{z,F}z} + A\,\mathrm{e}^{ik_{z,F}z} \\ \text{Support:} & D\,\mathrm{e}^{|k_{z,S}|\,(z+d_F/2)} \end{cases}$$

Idealized adlayer $n_A = n_C + c_A \frac{\mathrm{d}n}{\mathrm{d}c}$

Mass calculation $M = \mathrm{d}_A \frac{n_A - n_C}{\mathrm{d}n/\mathrm{d}c}$

6.3 Limitations

Above methods not usable for *diagnostic purpose* → NSB, LOD

Solution: diffractometric biosensors (Focal Molography)

7 Molecular Adsorption and Electron Transfer

7.1 Quantum Mechanics

Schrödinger equation $\hat{H}\psi(\vec{r}) = E\psi(\vec{r})$ $\quad$ where $\hat{H} = -\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + V$

1D potential well $\psi(x) = \sqrt{\frac{2}{L}}\sin(k_n x)$ $\quad$ for $0 \le x \le L$

where $k_n = \frac{n\pi}{L}$, $n \in \mathbb{N}$ and $E_n = \frac{\hbar^2 k_n^2}{2m}$

1D potential barrier (tunneling)

$$\psi(x) = \begin{cases} A\mathrm{e}^{ik'x} + B\mathrm{e}^{-ik'x} & x \le 0 \\ C\mathrm{e}^{ik''x} + D\mathrm{e}^{-ik''x} & \text{for } 0 \le x \le d \\ F\mathrm{e}^{ik'x} & \text{otw.} \end{cases}$$

where $k' = \sqrt{2mE}/\hbar$ and $k'' = \sqrt{2m(V_0 - E)}/\hbar$

transm./tunneling probability $T = \frac{F^*F}{A^*A} = B\mathrm{e}^{-\beta d}$ $\quad$ with $\beta = -2\sqrt{2m(V_0 - E)}/\hbar$

7.2 Electronic Transport through Molecules

Quantum conductance (1D) $G = G_0 \cdot T$ $\quad G_0 = \frac{2e^2}{h}$, $\quad$ in parallel: $G = N \cdot G_0$

Tunneling probability $T = B\mathrm{e}^{-\beta d} \propto \mathrm{e}^{-\beta d} = (\mathrm{e}^{-\beta_{NP} d})^N$

1D channel current $j = -(\mu_1 - \mu_2) e\, v\, \rho_E$ $\quad \rho_E = DOS = \frac{1}{\hbar\pi}\sqrt{\frac{2m}{E}}$

7.3 Atomic and Molecular Orbitals

Bond order: defined by difference (#electrons) divided by two

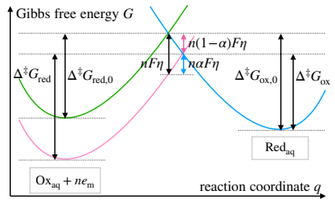
→ If the bond order is different from zero, then the bond is *stable*

7.4 Transition State Theory $A + B \to [AB] \to C + D$

Gibbs free energy $G = H - TS = U + pV - TS$

Ideal gas law $pV = n_{\mathrm{total}} RT$

7.4.1 Marcus Theory



See **section 9.1.1** (Butler-Volmer)

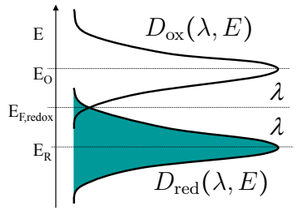
$k_{\mathrm{red}} = v_{\mathrm{red}}\, \mathrm{e}^{-\frac{\Delta^\ddagger G_{\mathrm{red}}}{RT}}$

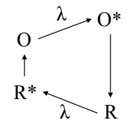
$k_{\mathrm{ox}} = v_{\mathrm{ox}}\, \mathrm{e}^{-\frac{\Delta^\ddagger G_{\mathrm{ox}}}{RT}}$

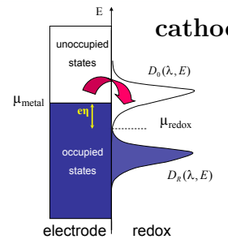
$\Delta^\ddagger G_{\mathrm{red}} = {}^\ddagger G - G_{\mathrm{ox,min}}$

$\Delta^\ddagger G_{\mathrm{ox}} = {}^\ddagger G - G_{\mathrm{red,min}}$

7.4.2 Gerischer's view







cathodic polariz. $i^+ \gg i^-$

8 Potentiometric Biosensors

8.1 Redox Reaction

oxidation: e^- donor (*anode*) $\quad$ **reduction**: e^- acceptor (*cathode*)

Chemical reaction $\mathrm{A}\cdot\mathrm{ox} + \mathrm{B}\cdot\mathrm{X} + \mathrm{z}\cdot\mathrm{e}^- \underset{k_{ox}}{\overset{k_{red}}{\rightleftharpoons}} \mathrm{C}\cdot\mathrm{red} + \mathrm{D}\cdot\mathrm{Y}$

reaction quotient $Q = \frac{a_{\mathrm{ox}}^{\mathrm{A}} \cdot a_{\mathrm{X}}^{\mathrm{B}}}{a_{\mathrm{red}}^{\mathrm{C}} \cdot a_{\mathrm{Y}}^{\mathrm{D}}}$ $\quad$ • $a_{\mathrm{ion}} = r_{\mathrm{ion}}[\mathrm{ion}]/1\,\mathrm{mol/l}$ • $a_{\mathrm{solids}} = 1$ • $a_{\mathrm{gas}} = p_{\mathrm{gas}}/1.013\,\mathrm{bar}$ • $[\mathrm{H_2O}] = 1$

Standard conditions $T = 1\,°\mathrm{C}$, $\quad p = 101.3\,\mathrm{kPa} = 1.013\,\mathrm{bar}$

"STP" $[\mathrm{ion}] \equiv c_{\mathrm{ion}} = 1\,\mathrm{mol/l}$, $\quad r = 1$ "activity coeff."

8.2 Electrochemistry

Dynamic equilibrium no net charge over time ($k_1 = k_{-1}$)

Chemical potential $\mu_j = \left(\frac{\partial G}{\partial n_j}\right)_{p,T,n'}$

Equilibrium $\mu_A = \mu_B$ $\quad$ general: $\sum_{\mathrm{prod}} v_j\mu_j = \sum_{\mathrm{react}} v_j\mu_j$

Chemical potential $\mu_{\mathrm{redox}} \equiv E_{F,\mathrm{redox}} \equiv E^*$

Equilibrium $\bar{\mu}_A = \bar{\mu}_B$ $\quad \bar{\mu}_j = \mu_j^0 + z_j F\Delta\phi$

Contact potential $\Delta\phi = V_{\mathrm{in}} - V_{\mathrm{out}} = \frac{k_B T}{Q}\ln\left(\frac{[C]_{\mathrm{out}}}{[C]_{\mathrm{in}}}\right)$

$= -\frac{\Delta_r G}{zF} = \frac{1}{zF}\left(\sum_{\mathrm{ox}} v_j\mu_j - \sum_{\mathrm{red}} v_j\mu_j + z\mu_e\right)$

Nernst equation, $\Delta\phi$ $\quad E = E^0 - \frac{59\,\mathrm{mV}}{z}\log_{10} Q$ $\quad \frac{RT}{zF}\ln Q = \frac{2.303RT}{zF}\log_{10} Q$

$E_{\mathrm{cell}} = E_1 - E_2$ $\quad$ or $\quad$ let $E_1 \overset{!}{=} E_2 \to \mathrm{pH} = \ldots$

8.3 Ion-selective Electrodes (ISE)

Sensor is seperated to solution through a H^+ permeable glass.

pH electrode $\mathrm{pH} = -\log_{10} a_{\mathrm{H^+}} = \frac{K' - \Delta\phi}{0.059\,\mathrm{V}}$

8.4 Bioenzymatic Electrodes

Find out how much of an enzyme was initially present

$\mathrm{substrate} + \mathrm{X} + \ldots \xrightarrow{\text{enzyme}} \mathrm{product} + \mathrm{Y} + \ldots$

Detect **"a lot"** X

↔ **"very few"** substrate was initially present

9 Amperometric Sensors

9.1 Electrochemistry

Overpotential $\eta \equiv \Delta\phi_{\mathrm{appl}} - \Delta\phi^0 \equiv \phi_s - \phi_m \equiv E - E^0$ $\quad (E^0$ via Nernst)

$\eta = 0$ $\quad$ (**equilibrium** i.e. no net current)

Faraday's law of electrolysis *1st:* $n \propto Q$, $\quad$ *2nd:* (equiv. weigth) $W_{\mathrm{eq}} = M/z$

$m = \frac{Q}{zF}M = \frac{It}{zF}M = n \cdot M$ $\quad [m] = \mathrm{g}$, $[n] = \mathrm{mol}$

9.1.1 Butler-Volmer equation Effect of η on barrier heigth G

Let ϕ_s: potential of ions in solution

ϕ_m: potential of e^- in (metallic) electrode

z: valency of oxidized species

n: #of transferred e^-

$z - n$: valency of reduced species

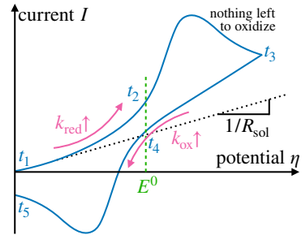
transfer coeff. α $\quad nF\eta = n(1-\alpha)F\eta + n\alpha F\eta$

$f \equiv \frac{F}{RT} \overset{@298\,\mathrm{K}}{=} \frac{1}{25.69\,\mathrm{mV}}$ $\quad k_{\mathrm{red}} = k_0\, \mathrm{e}^{-n\alpha f\eta}$, $\quad k_{\mathrm{ox}} = k_0\, \mathrm{e}^{n(1-\alpha)f\eta}$ $\quad$ @equi. ($\eta = 0$) $k_0 \equiv k_{\mathrm{red}} = k_{\mathrm{ox}}$

The Butler-Vomer equation relates what we measure (the **current**) with what we would like to determine (the **concentration** of an analyte):

Butler-Volmer $j = \underbrace{nFAk_0}_{j_0}\left(C_{\mathrm{ox}}(0,t)\,\mathrm{e}^{-\alpha nf\eta} - C_{\mathrm{red}}(0,t)\,\mathrm{e}^{(1-\alpha)nf\eta}\right)$

9.2 Cyclic Voltammetry



Offers information on the mechanism of the ec reactions occuring at an electrode.

- May be irreversible
  → #upper peaks ≠ #lower peaks
  → irreversible reaction fast
- Sweep rate
  → irreversible reaction slow

Water electrolysis High potentials/volt's

→ affects I (disturbance, bad)

9.3 Amperometric Sensors

9.3.1 Clark (Oxygen) Electrode → Measure O_2

- test solution $\xrightarrow[\text{membrane}]{\mathrm{O_2\ passes}}$ Pt cathode $\xrightarrow{\text{reduction}}$ current
- Ag anode is in KCl solution (→ "enough" Cl^- for oxidation)

9.3.2 1st and 2nd Generation

analyte of interest … $\underbrace{\text{redox enzyme}}_{\text{catalize}}$ $\underbrace{\ldots}_{\text{P1}}$ $\underbrace{\text{mediator}}_{\text{P2}}$ … Clark Electrode

problem P1 (1st) O_2 may be consumed, when not measured

problem P2 (2nd) need to have "enough" of them

Mediator needs to be • reversible • not toxic • no side reactions

9.3.3 3rd Generation

Immobilization/fixation of a redox enzyme on electrode surface

→ free-diffusing redox mediators are not necessary

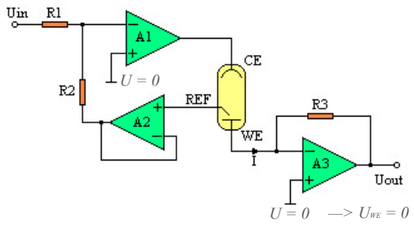
→ in vivo measurments allowed since immobilized

problem P3 Efficient electron transfer (Marcus theory)

→ may be overcome by mediators

→ minimize ET distance

9.4 Three Electrode Cell



$$\frac{U_{\text{REF}}}{R_2} = -\frac{U_{\text{in}}}{R_1}$$
$$I = \frac{U_{\text{out}}}{R_3}$$

10 Membranes and Transport

10.1 Diffusion

Concentration $c(x,t)\ [\text{mol/l} = \text{mol}/10^3\,\text{cm}^3]$

Flux $\phi(x,t)\ [\text{mol}/\text{cm}^2\,\text{s}]$

Fick's law $\phi(x,t) = -D_n \frac{\partial c_n(x,t)}{\partial x} = \phi_0$ in **steady state**

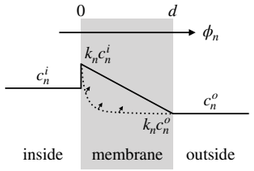
D_n: diffusion coeff $[\text{cm}^2/\text{s}]$

Continuity eq. $-\frac{\partial \phi_n(x,t)}{\partial x} = \frac{\partial c_n(x,t)}{\partial t}$

e.g. **Delta initial condition** $c(x,t)|_{t=0} = n_0\delta(x)$

$$c(x,t) = \frac{n_0}{\sqrt{2\pi\sigma}}\, e^{-x^2/\sigma^2}, \quad \sigma = \sqrt{2D_n t}, \quad t_{1/2} = \frac{1}{D_n}\, x_{1/2}^2$$

10.1.1 Diffusion across membrane



$\phi_n(t) = P_n\,(c_n^i(t) - c_n^o(t))$
with permeability: $P_n = \frac{D_n k_n}{d}$

state space (inside membrane)
after $\tau_{\text{SS}} = \frac{d^2}{\pi^2 D_n}$

10.2 Osmosis

Osmotic (back-)pressure $\pi(x,t) = RT \sum_n c_n(x,t)$ $\quad c_\Sigma(x,t)$: osmolarity/ total conc.

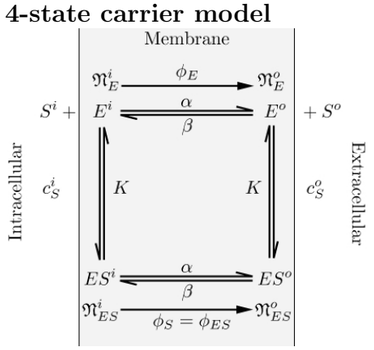
Zero pressure if $\pi_i - \pi_o = 0$

Volume of the cell $v_c(\infty) = v_c' + \frac{N_\Sigma^i}{c_\Sigma^o}$ $\quad v_c'$: non-water volume

10.3 Carrier mediated Transport

Hints for their existance
- Saturation of solute trsp. (↯ Fick's law)
- Competitive inhibiton
- Structure specificy

4-state carrier model



1. Conservation
   $N_E^i + N_E^o + N_{ES}^i + N_{ES}^o = N_{ET}$
2. Fast binding (always steady state)
   $K = \frac{c_S^i N_E^i}{N_{ES}^i} = \frac{c_S^o N_E^o}{N_{ES}^o}$
3. Translocation (char. by flux)
   $\phi_{E(S)} = \alpha N_{E(S)}^i - \beta N_{E(S)}^o$
4. Zero net flux
   $\phi_E + \underbrace{\phi_{ES}}_{\phi_S} = 0$

Net flux out of cell $\phi_S = \frac{\alpha\beta}{\alpha+\beta} N_{ET}\left(\frac{c_s^i}{c_s^i+K} - \frac{c_s^o}{c_s^o+K}\right)$ $\quad (\phi_S)_{\max} = \frac{\alpha\beta}{\alpha+\beta} N_{ET}$

10.4 Ion Transport

Continuity eq. $\frac{\partial J_n(x,t)}{\partial x} = -z_n F \frac{\partial c_n(x,t)}{\partial t}$ $\quad z_n$: valency

Poisson eq. $\frac{\partial^2 \psi(x,t)}{\partial x^2} = -\frac{1}{\epsilon}\sum_n z_n F c_n(x,t)$

i.e. electric field potential = local charge density

Nernst-Planck eq. u_n: molar mobility

$$\underbrace{J_n(x,t)}_{\text{current}} = -\underbrace{z_n F D_n \frac{\partial c_n(x,t)}{\partial x}}_{\text{diffusion of charge}} - \underbrace{u_n z_n^2 F^2 c_n(x,t)\frac{\partial \psi(x,t)}{\partial x}}_{\text{mobility of charge in E-field}}$$

Debye Length "depletion width" $\lambda_D \simeq 1\,\text{nm}$ $\quad$ within $\tau_r \simeq 1\,\text{ns}$

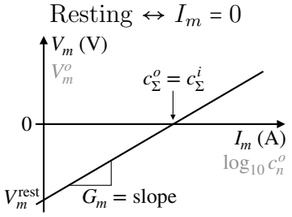
Nernst Equi. Pot. $V_n = \frac{59\,\text{mV}}{z_n}\log_{10} Q$ with $Q = \frac{c_s^o}{c_s^i}$, $\quad \frac{D_n}{u_n} = RT$

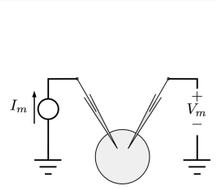
$V_{\text{K}} \simeq -75\,\text{mV}$, $\quad V_{\text{Na}} \simeq 55\,\text{mV}$

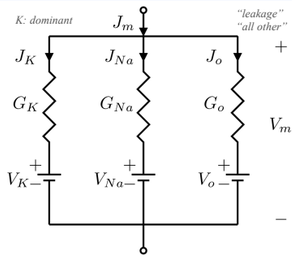
Donnan equilibrium $V_m = V_n \implies I_n = 0$ $\quad \left(\frac{c_{n_i}^o}{c_n^i}\right)^{\pm\frac{1}{z_n}} = e^{\frac{FV_m}{RT}}$ $\quad$ +: anodic

10.4.1 Resting (membrane) potential

Resting $\leftrightarrow I_m = 0$







3-ion model permeable to K, Na and some others

Resting potential $V_m^{\text{rest}} = \frac{V_K G_K}{G_{\text{tot}}} + \frac{V_{\text{Na}} G_{\text{Na}}}{G_{\text{tot}}} + \frac{V_o G_o}{G_{\text{tot}}}$ $\quad G_{\text{tot}} \equiv G_m$ = slope

$J_m = J_{\text{K}} + J_{\text{Na}} + J_o = 0$ $\quad$ where $J_n \neq 0$ (only momentarly rest)

10.5 Active Transport

Rest is only momentarly → maintain conc. gradients with *active transport*.
- Ions moving against its conc. gradient → ATP needed
- Active pumps for $J_m = 0$ (rest) and $J_{n,\text{in}} + J_{n,\text{out}} = 0$ (quasi equilibrium).

11 Action Potential & Hodgkin-Huxley Model

11.1 Current Clamp

Fix current I_m and measure membrane potential

→ Good for observing AP (since voltage can change)

11.2 Voltage Clamp

Fix voltage V_m and measure the current

→ Good for studying membrane channel proteins (ion transport)

11.3 2-state ion channel model

two states close $\underset{\beta}{\overset{\alpha}{\rightleftharpoons}}$ open $\quad$ with gate charge Q

#open states $\frac{\text{d}n(t)}{\text{d}t} = \alpha(\mathcal{N} - n(t)) - \beta n(t)$

Prob. being open $x(t) \simeq \frac{n(t)}{\mathcal{N}} = x_\infty + (x_0 - x_\infty)\, e^{-t/\tau_x}$

$x_\infty = \frac{\alpha}{\alpha+\beta}$, $\quad \tau_x = \frac{1}{\alpha+\beta}$

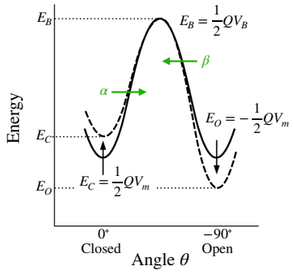
Boltzmann law

$\alpha = A\, e^{(E_c - E_B)/kT}$

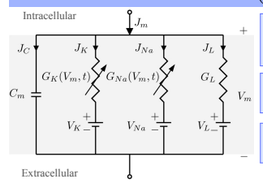
$\beta = A\, e^{(E_o - E_B)/kT}$

$x_\infty = \frac{1}{1+\beta/\alpha} = \frac{1}{1+e^{-QV_m/kT}}$

$\tau_x = \frac{1}{A\, e^{-\frac{1}{2}QV_B/kT}} \cdots \frac{1}{e^{\frac{1}{2}QV_m/kT} + e^{-\frac{1}{2}QV_m/kT}}$



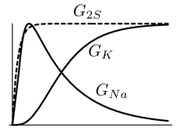
11.4 Circuit Model (Conductance) $\quad V_{\text{K}} \simeq -75\,\text{mV}$, $\quad V_{\text{Na}} \simeq +55\,\text{mV}$



$J_C = C_m \frac{\text{d}V_m(t)}{\text{d}t}$ $\quad$ $J_m = C_m \frac{\text{d}V_m}{\text{d}t} + G_m(V_m - V_m^{\text{rest}})$

$J_{\text{K}} = G_{\text{K}}\,(V_m - V_{\text{K}})$ always > 0, we don't fall below ($V_m > V_{\text{K}}$)

$J_{\text{Na}} = G_{\text{Na}}\,(V_m - V_{\text{Na}})$ $\quad \begin{cases} > 0 & \text{if } V_m - V_{\text{Na}} > 0 \\ < 0 & \text{if otw.} \end{cases}$



- slow onset; decays slow
- K^+ inactivates Na^+ channels

→ **Refractoriness** (kurzzeitige AP Resistenz)
(Na^+ channels are still inactivate, $h \sim 0$)

11.5 Multiple states gate

open $n_Q(V_m, t)$ $\quad$ big/small Q → fast/slow dynamics

Let's write $\boxed{h} = n_{-Q}$ "slow", $\quad \boxed{m} = n_{2Q}$ "fast", $\quad \boxed{n} = n_Q$

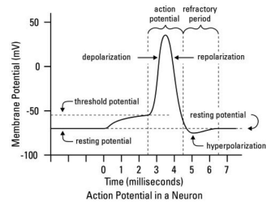
The *lower* T, the *bigger* the difference btw them.

11.6 Hodkin-Huxley (HH) model

Experimentally you may fit the K^+ and Na^+ current into

Conductance $G_{\text{K}}(V_m, t) = \overline{G}_{\text{K}}\, n^4$

$G_{\text{Na}}(V_m, t) = \overline{G}_{\text{Na}}\, m^3 h$



Four regimes during AP:

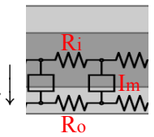
II *De*polarization: Na^+ gate opens → Na^+ in

III *Re*polarization: K^+ gate opens → K^+ out

IV *Hyper*polarization:
Active Na^+–K^+–ion pump ($Na^+ \leftrightarrow K^+$)

11.7 Decrement-free conduction

11.7.1 Core – Conductor Model



$R_i = r_i\,\text{d}z$ ← dz: per unit length

$R_o = r_o\,\text{d}z$

$I_m = k_m\,\text{d}z$ ← may use HH-model

Core–Conductor eq. $\frac{\partial^2 V_m(z,t)}{\partial z^2} = (r_o + r_i)K_m(z,t) - \overbrace{r_o K_e(z,t)}^{\text{if } = 0 \to I_o = -I_i}$

wave eq. $= \frac{1}{v^2}\frac{\partial^2 V_m(z,t)}{\partial t^2}$ $\quad$ with $v = \frac{W}{\Delta t}$

$\overset{K_e=0}{\implies}$ $\boxed{v \simeq \frac{K_m a}{2\rho_i}}$ $\quad$ for $r_i \gg r_o$ $\quad$ i.e. $v \propto \sqrt{a}$

11.7.2 Cable model – Core–Conductor with HH-model inside

Linearize (1st order) → timescale for membrane voltage changes $\tau_m = \frac{C_m}{G_m}$

Cable equation Let $v_m = V_m + V_m^{\text{rest}}$

$$v_m + \underbrace{\tau_m \frac{\partial v_m}{\partial t}}_{=0 \text{ time indep.}} - \lambda_C^2 \frac{\partial^2 v_m}{\partial z^2} = r_o \lambda_C^2 K_e \qquad \boxed{\lambda_C = \frac{1}{\sqrt{g_m(r_o + r_i)}}}$$

11.7.3 Saltatory Conduction Hypothesis

→ explain discrete manner in steps

velocity ~ axon diameter D $\quad$ total delay ~ N(#nodes) ~ (axon length)/L

$\xrightarrow{L \sim D}$ velocity ~ $\frac{\text{total delay}}{\text{axon length}} \sim D$

